

Inland Penetration of Lake Effect Snowfall in Superior Region

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Lake effect snowfall (LES) events in the Michigan peninsula are collected. Inland penetrance is calculated and synoptic variables are either directly obtained or engineered, and ran through two separate machine learning (ML) models. Results suggest that kinematic variables may play a larger role in inland penetration than previously expected. ML models need to be painstakingly fine-tuned before they can become a viable option for analysis of mesoscale events like LES.

Background

- **Objective:** Assess ability of ML models to predict inland penetration of LES. Uncover new synoptic-scale variables to study that may drive or restrain inland penetrance.
- **LES Fundamentals:** Driven by cold air masses moving for long distances over relatively warm bodies of water, forcing parcels to rise, condense, and precipitate intensely as snowfall.
- **Typical Variables:**
 - Delta-T, or the change in temperature from surface to cloud level – historically the most influential variable
 - Boundary layer height (BLH)
 - Existence of multi-lake connection (MLC)

Table 1: Correlation to inland extent of variables (bold for significance) from Villani et. al. (2017) study on inland penetrance

Parameter	Correlation to inland extent (observed ALY and BUF soundings)
Existence of MLC	0.64* (point biserial correlation)
Mean ML wind speed	0.29
Surface wind speed	0.26
Wind speed at top of ML	0.23
Mean ML wind direction	0.17
Surface wind direction	0.11
Wind direction at top of ML	0.10
0–1-km bulk shear	0.06
ML bulk shear	0.02
Capping inversion height (or ML depth)	0.01
LES band width	−0.12
0–3-km bulk shear	−0.20
Surface dewpoint depression	−0.20
Maximum ML dewpoint depression	−0.30
Lake – air ΔT at 700 hPa (°C)	−0.37
Lake – air ΔT at 850 hPa (°C)	−0.56

- Villani et. al. (2017) developed a 14-variable linear model that explained ~75% of the variance in land penetration
- Eipper et. al. (2018) further studied synoptic variables’ impact on inland penetrance, finding importance of instability
- Linear models are unable to understand complex boundary conditions, while ML models can, as well as uncovering new variables’ feature importance

Methods & Data

- **NOAA NCEI Database (Event Ledger)**
 - Pulled winter weather advisories, warnings, and other snow events for 8 different counties in Michigan’s peninsula region (Fig. 2), from 2018 to present
 - Filtered the event description text for “lake effect” or “lake enhanced”, and multi-day events reduced to a singular item
- **Local Storm Reports (Ground-based observations)**
 - Used the Iowa Environmental Mesonet (IEM) to pull the data for the 8 counties used for the study
 - Threshold of > 1 inch of reported accumulation during LES events. Maximum reported distance meeting the threshold from the coast during any event was taken as the inland penetrance
 - Used as the target (Y) for the ML model
- **Era5 Reanalysis (Synoptic-scale variables)**
 - Utilized Copernicus Era5 data to construct 3D atmospheric grid
 - Instead of daily averages, 48-hour windows were scanned to extract the peak convective hour
 - Derived 8 features to map kinematics and thermodynamics, including low-level lapse rates, synoptic lift, and vertical directional shear (Fig. 3)
- **ML Algorithms**
 - Trained Random Forest and XGBoost models to capture non-linear, threshold-based atmospheric interactions
 - Used Leave-One-Out Cross-Validation (LOOCV) to prevent overfitting (given the dataset was small, only 134 final events)
 - Isolated specific physical drivers of inland band survival

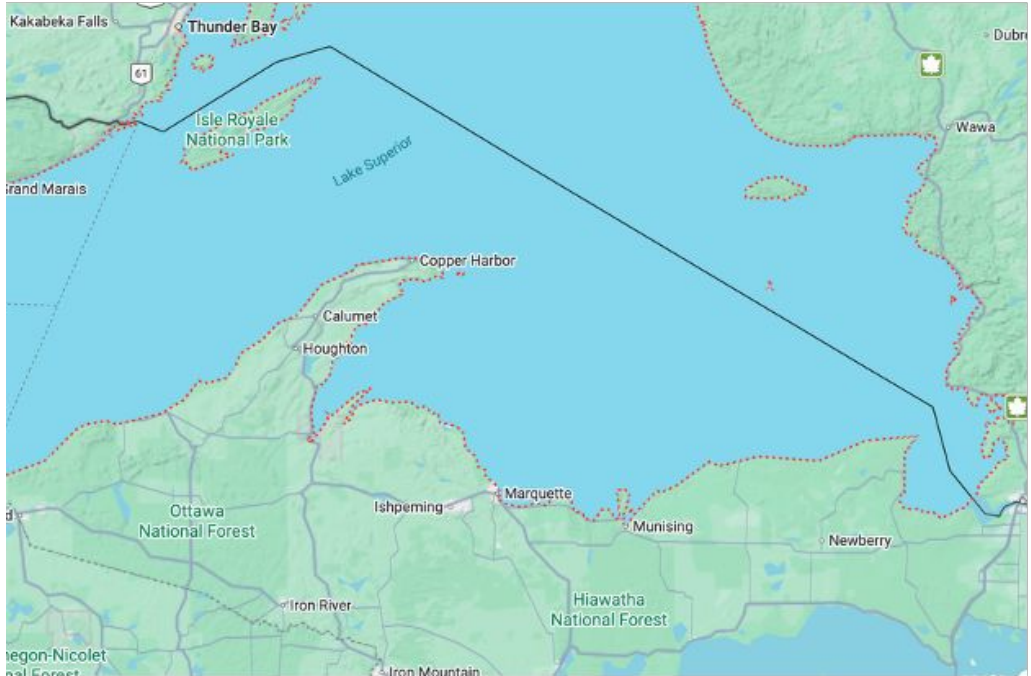


Figure 1: Area of analysis: Upper peninsula of Michigan, with fetch vectors pointing SW to S across Lake Superior

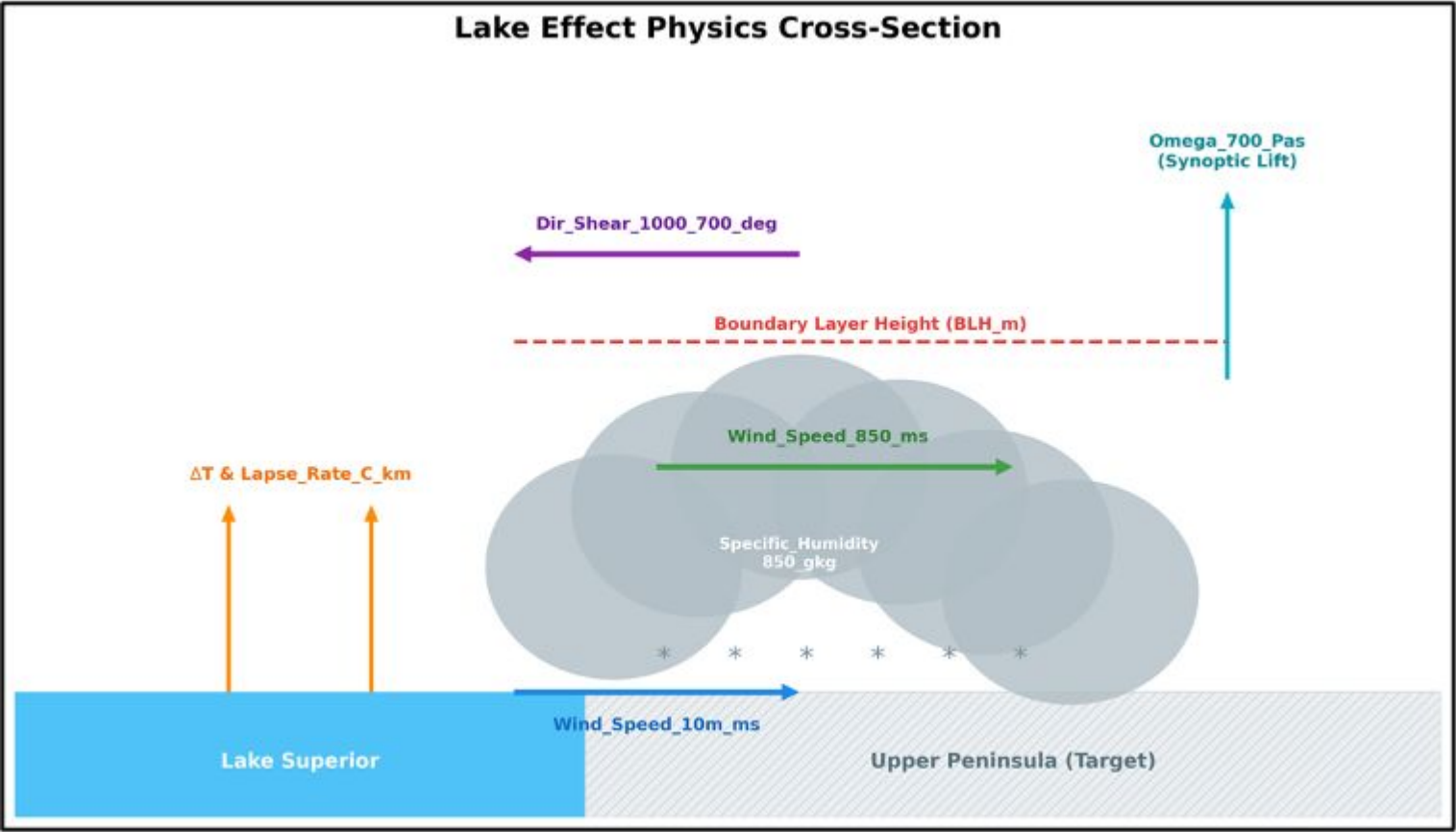


Figure 2: Diagram of several variables engineered for analysis using ML models. Variables overlaid on graphic of typical LES event.

Results

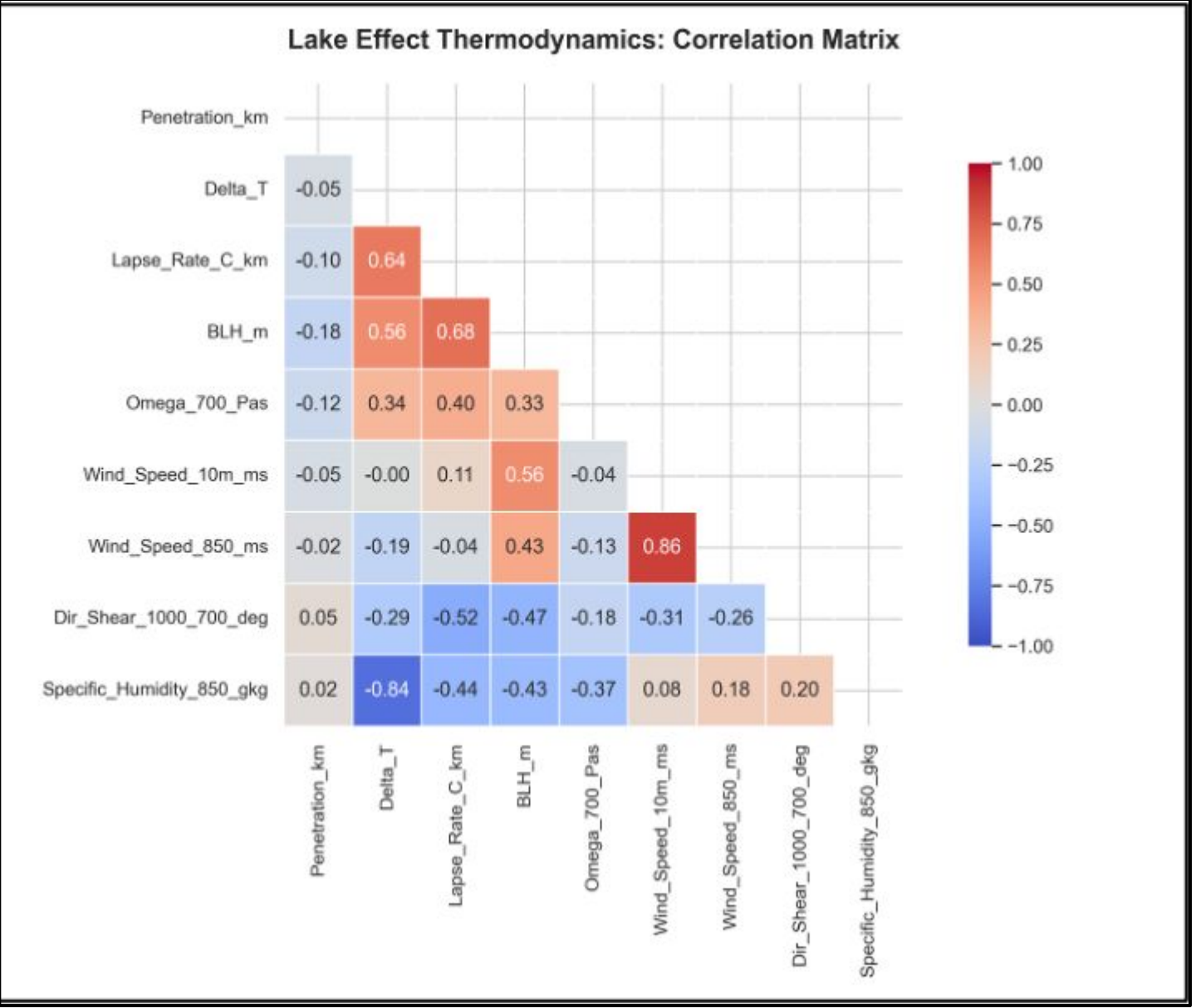


Figure 3: Heatplot for variables incorporated into ML models. Expected relationships are observed.

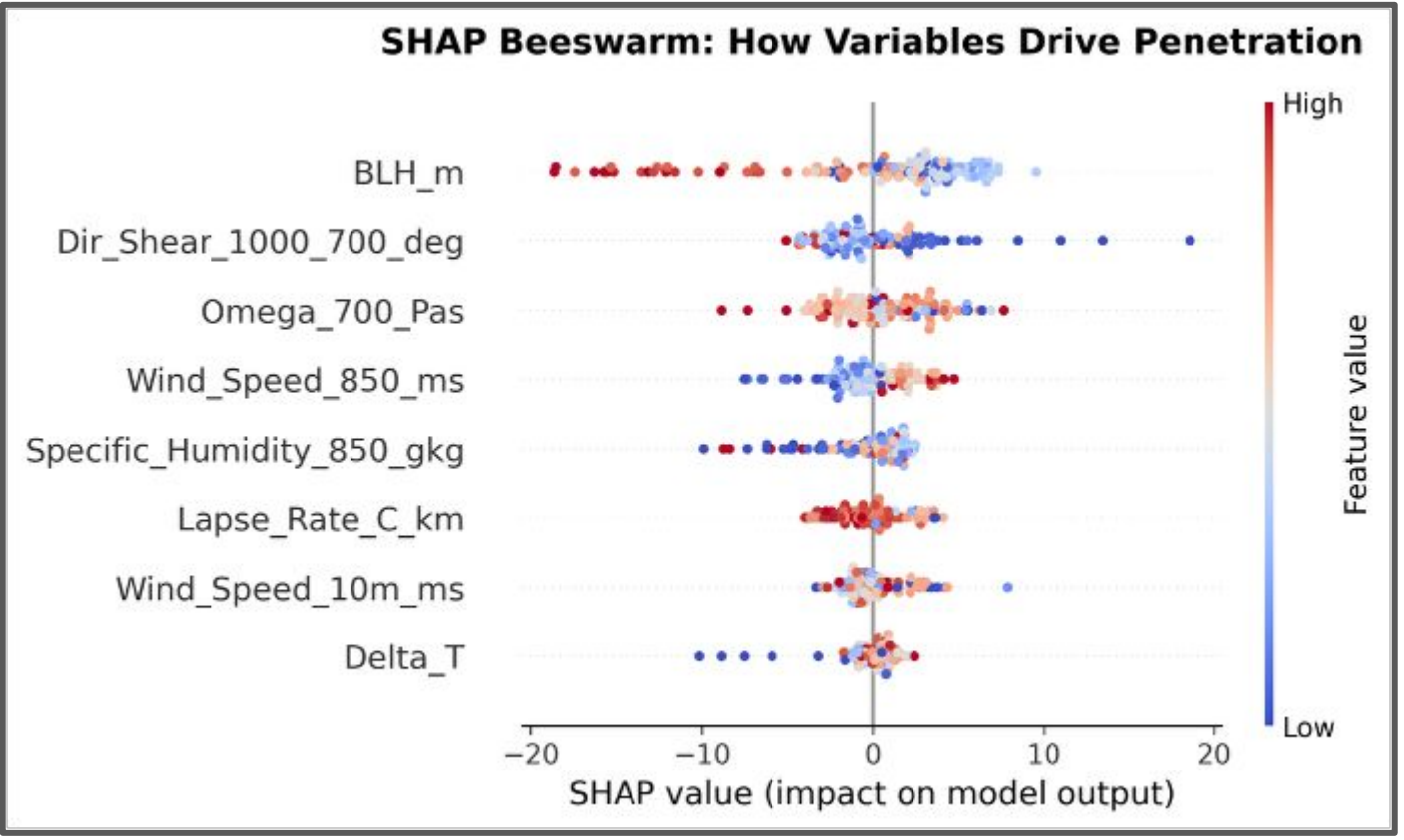
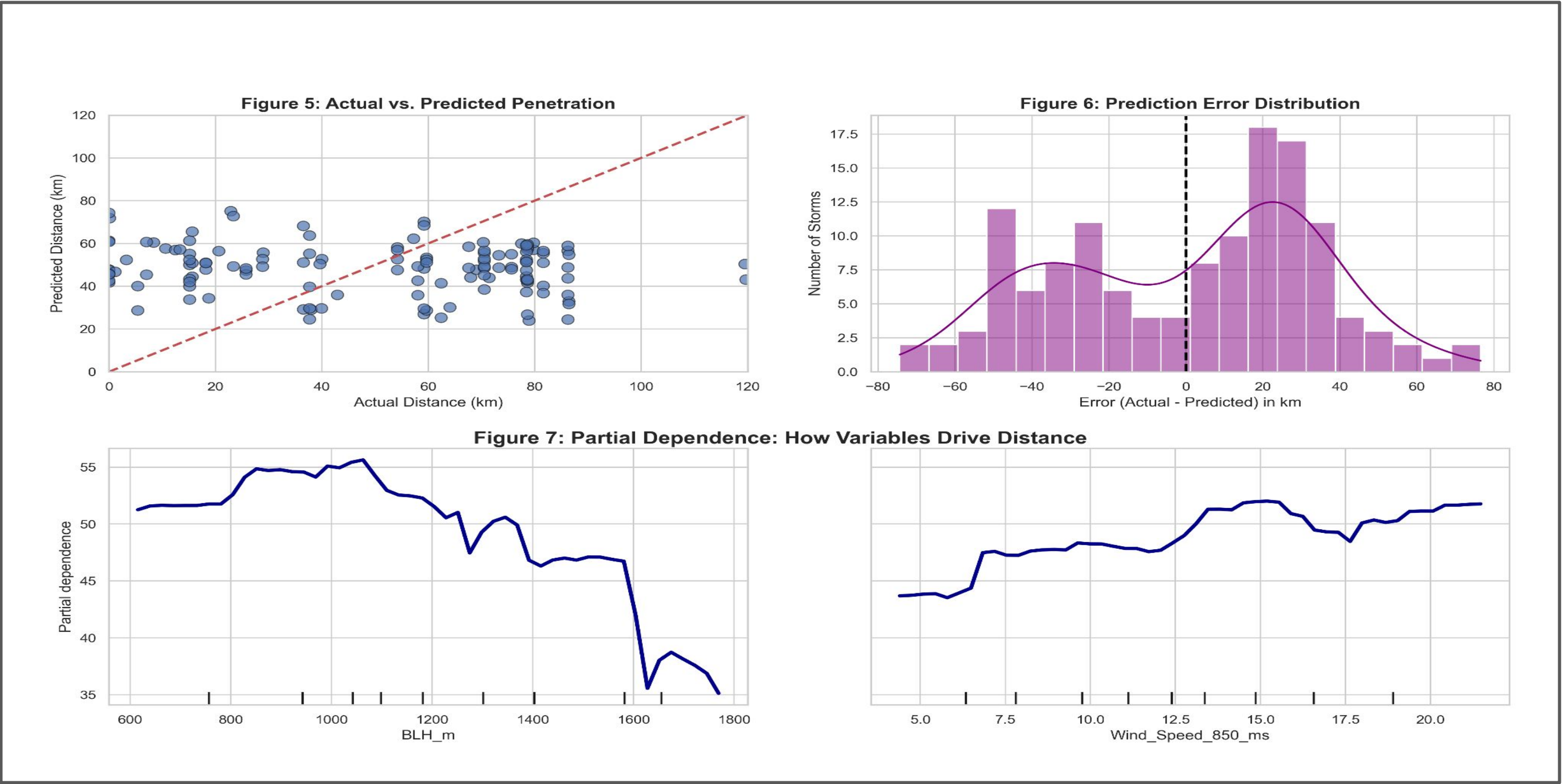


Figure 4: Feature importance plot with directionality (SHAP plot) for Random Forest model.



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Discussion & Conclusion

- The dataset appears to be valid, with the expected correlations between variables (e.g. 10-m wind speed strongly positively correlated with 850-hPa wind speed)
- Mean average error (MAE) was high for both models (~30km), likely driven by the following limitations: Dropped variables like latent heat flux from the Era5 data due to mismatches in how it was downloaded; could only calculate proxies for inland penetrance (and storm direction), since ground-based observational data was the primary method; Era5 data is course and it is generally difficult to map synoptic-scale data to mesoscale events (like LES); the area of interest was underpopulated and thus subject to human bias
- Kinematic variables like wind shear and subsidence appear to play more important roles than historically important thermodynamic variables like Delta-T under the machine learning analysis
- The high MAE points to complex synoptic influence, at least in this region, as compared to the more typically-studied Lake Ottawa regions
- There was some correlation between ML output and Villani et. al.’s study, but the magnitude of differences suggests it is not feasible to extend a linear analysis from one synoptic region to another